

Experiment No. : 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: VS code, Web browser

3. Theory: A billing calculator demonstrates var, let, const, template literals, and destructuring by processing user inputs, assigning constants (like tax), calculating totals (using updated variables), and dynamically formatting the result.

4. Output:

Billing Calculator

Item Name:

Wireless Mouse

Quantity:

2

Unit Price (\$):

25.50

Tax Rate (%):

8

Test Inline JS (Check Console)

Generate Invoice

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Tech Solutions Inc.

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Invoice #: 1001

Item: Wireless Mouse

Qty: 2 x \$25.50

Subtotal: \$51.00

Tax (8%): \$4.08

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TOTAL DUE: \$55.08

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5.Result/Conclusion

The billing calculator was successfully developed using JavaScript to process user inputs and generate a dynamic billing receipt. The experiment demonstrated the effective use of modern JavaScript (ES6+) features, including **var**, **let**, **const**, **template literals**, and **object destructuring**, while creating an interactive web application.

Key Learning Outcomes:

- **Variable Scoping:** Understood the differences between

- **var, let, and const.** const was used for values that remain unchanged, let for variables requiring reassignment during calculations, and var to understand legacy function-scoped variable behavior.
- **Data Manipulation:** Applied **object destructuring** to efficiently extract user input values, making the code cleaner, more readable, and easier to maintain.
- **Dynamic Rendering:** Used **template literals** to generate well-formatted, multi-line billing receipts with embedded variable values, improving the presentation of the output.
- **Interactive User Input:** Implemented user input handling and real-time calculations to compute the total bill accurately based on the entered values.

Conclusion:

This experiment successfully demonstrated how modern JavaScript features can be applied to build a simple, interactive billing calculator. It strengthened the understanding of variable declarations, string formatting, object destructuring, and DOM manipulation, while highlighting how ES6+ features improve code readability, maintainability, and overall development efficiency.